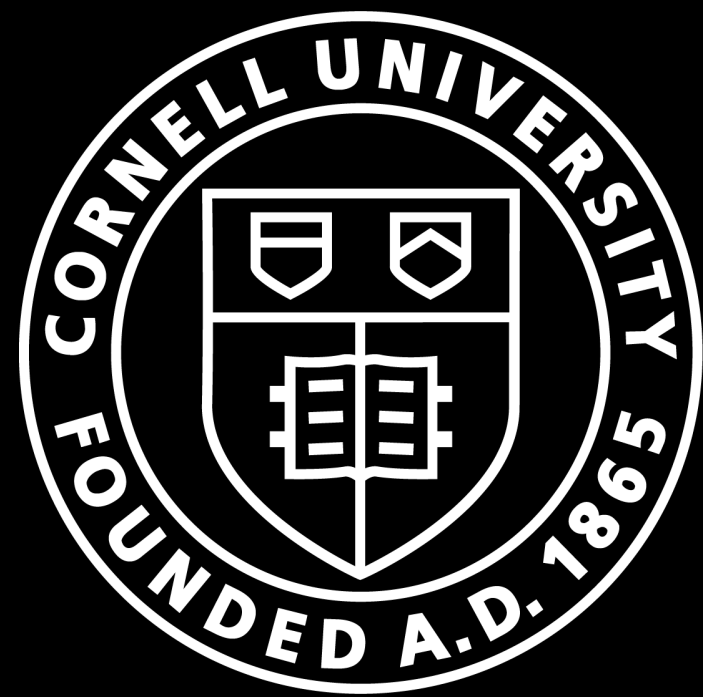




BSTModelKit.jl: Building Biochemical Systems Theory Models of Individualized Coagulation Dynamics During Pregnancy

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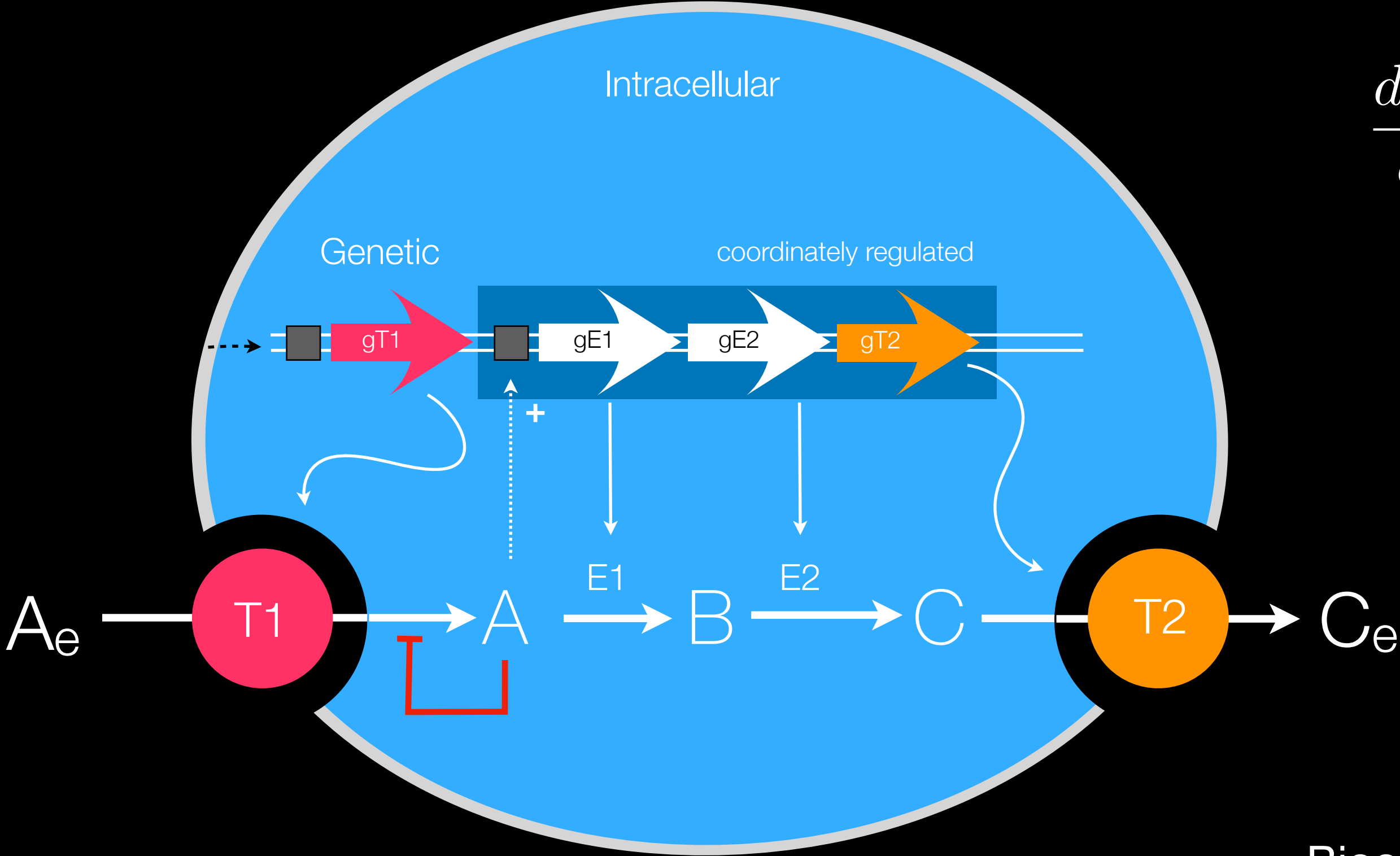
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BSTModelKit.jl is a lightweight package that transforms model specification files into Biochemical Systems Theory (BST) models of metabolic or signal transduction networks and solves them using `DifferentialEquations.jl`

GitHub: <https://github.com/varnerlab/BSTModelKit.jl>

Background: What is a BST model?



Traditional kinetic model concentration balances

$$\frac{dX_i}{dt} = \underbrace{\sum_{j \in \mathcal{P}} \sigma_{ij} \hat{r}_j(X, \kappa)}_{V_i} - \underbrace{\sum_{j \in \mathcal{R}} \sigma_{ij} \hat{r}_j(X, \kappa)}_{V_{-i}} \quad i = 1, \dots, n$$

Power-law approximation of V_i and V_{-i}

Taylor expand V_i and V_{-i} in log space (keep up to first-order terms)

Transform back into rectangular coordinates

Biochemical Systems Theory (BST) kinetic concentration balances

(S-system representation)

$$\frac{dX_i}{dt} = \alpha_i \prod_{j \in \mathcal{P}} X_j^{g_{ij}} - \beta_i \prod_{j \in \mathcal{R}} X_j^{h_{ij}} \quad i = 1, \dots, n$$

Example: Coagulation.bst

```
// s-law model for extrinsic and intrinsic coagulation cascade -

#static::start
TFPI,AT,TF,TM,PC,TRAUMA,SURFACE
#static::end

#dynamic::start
FII,FVII,FV,FX,FVIII,FXI,FXII,FIIa,FVIIa,FVa,FXa,FVIIIa,FXa,FXIa,FXIIa,FIIa_inactive
#dynamic::end

#structure::start
  // name,{reactants},{products}
  // 1. name: reaction name (used in rate section)
  // 2. {reactants}: list of dynamic species acting as a reactants
  // 3. {products}: list of dynamic species acting as a reactants
  r1::FVII --> FVIIa
  r2::FVIII --> FVIIIa
  r3::FIX --> FIXa
  r4::FV --> FVa
  r5::FX --> FXa
  r6::FII --> FIIa
  r7::FXI --> FXIa
  r8::FXII --> FXIIa
  r9::FIIa --> FIIa_inactive
#structure::end

#rate::start
  // name,{factor_1,factor_2,...}
  // 1: name = reaction name (must be in the #pragma::structure section)
  // 2: list of factor symbols (can be both static and dynamic, order does not matter)
  r1::{FVII,TRAUMA,TF,TFPI}
  r2::{FVIII,FIIa}
  r3::{FIX,FVIIa,FXIa}
  r4::{FV,FIIa}
  r5::{FX,FVIIa}
  r6::{FII,FXa,FVa}
  r7::{FXI,FXIIa}
  r8::{FXII,SURFACE}
  r9::{FIIa,AT}
#rate::end
```

Model specifications: Flat file, JSON, and (working on) SBML

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BSTModelKit.jl

model = build(path_to_model_file);

Key exported functions

```
function evaluate(model::BSTModel; tspan::Tuple{Float64,Float64} = (0.0,20.0), Δt::Float64 = 0.01,
  input::Union{Nothing,Function} = nothing)::Tuple{Array{Float64,1}, Array{Float64,2}}

function steadystate(model::BSTModel; tspan::Tuple{Float64,Float64} = (0.0,20.0), Δt::Float64 = 0.01,
  input::Union{Nothing,Function} = nothing)::Array{Float64,1}

function morris(performance::Function, L::Array{Float64,1}, U::Array{Float64,1};
  number_of_samples::Int64 = 1000)::Array{Float64,2}
```

BSTModel type holds

```
mutable struct BSTModel <: AbstractBSTModel

  # data -
  number_of_dynamic_states::Int64
  number_of_static_states::Int64
  list_of_dynamic_species::Array{String,1}
  list_of_static_species::Array{String,1}
  list_of_reactions::Array{String,1}
  total_species_list::Array{String,1}
  static_factors_array::Array{Float64,1}
  initial_condition_array::Array{Float64,1}
  S::Array{Float64,2}
  G::Array{Float64,2}
  α::Array{Float64,1}

  # metadata -
  author::String
  version::String
  date::String
  description::String

  # constructor -
  BSTModel() = new()

end
```




U.S. Maternal Mortality Hits Highest Level Since 1965

Black mothers are the most affected, 2021 data show.

By Sarah Toy

Wall Street Journal, March 16, 2023

U.S. Maternal Mortality Hits Highest Level Since 1965

Between 2018 and 2019, the rate increased from 17.4 maternal deaths per 100 000 live births to 20.1. It then rose to 23.8 in 2020 and 32.9 in 2021. There were 1205 maternal deaths in 2021 compared with 658 in 2018.

- The coagulation and fibrinolytic proteomes are significantly altered in pregnancy, often leading to hypercoagulability (increased clotting potential)

Pregnant and postpartum women have a 4-5 times higher risk of venous thromboembolism than non-pregnant women.

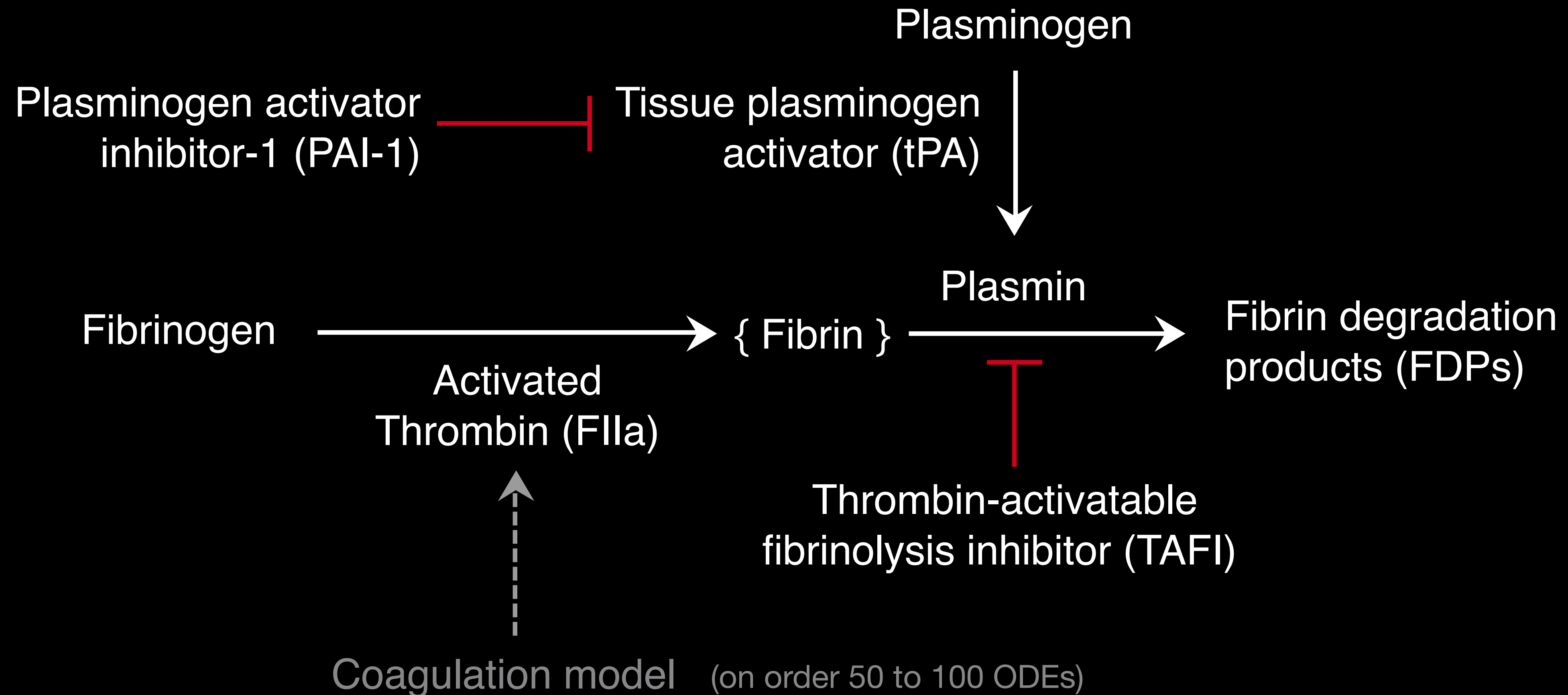
Between 2017 and 2019, thrombotic embolisms caused 8.7% of pregnancy-related deaths, while hemorrhage caused 13.7%

Harris E. US Maternal Mortality Continues to Worsen. *JAMA*. 2023;329(15):1248. doi:10.1001/jama.2023.5254

Trost SL, Beauregard J, Njie F, et al. Pregnancy-Related Deaths: Data from Maternal Mortality Review Committees in 36 US States, 2017-2019. Atlanta, GA: Centers for Disease Control and Prevention, US Department of Health and Human Services; 2022.

Grouzi, E., Pouliakis, A., Aktypi, A. et al. Pregnancy and thrombosis risk for women without a history of thrombotic events: a retrospective study of the real risks. *Thrombosis J* 20, 60 (2022). <https://doi.org/10.1186/s12959-022-00419-6>

We developed a Biochemical Systems Theory (BST) model of Thrombin generation and Fibrinolysis. BST models are low-order alternatives to traditional kinetic models



Hockin MF, Jones KC, Everse SJ, Mann KG. A model for the stoichiometric regulation of blood coagulation. J Biol Chem. 2002 May 24;277(21):18322-33. doi: 10.1074/jbc.M201173200. Epub 2002 Mar 13. PMID: 11893748.

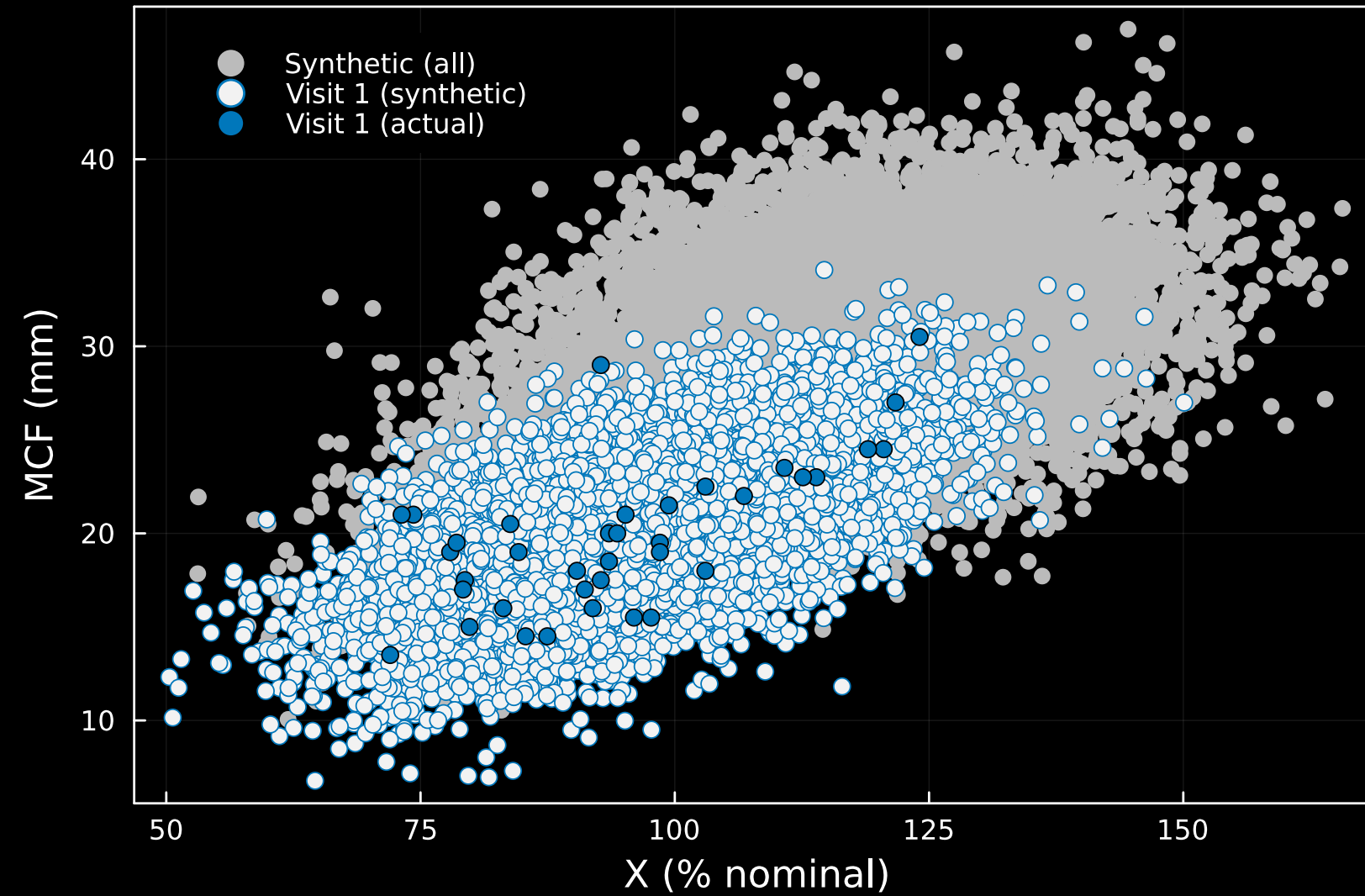
Luan D, Zai M, Varner J. Computationally derived points of fragility of a human cascade are consistent with current therapeutic strategies. PLoS Comput Biol. 2007 Jul;3(7):e142. doi: 10.1371/journal.pcbi.0030142. PMID: 17658944; PMCID: PMC1924874.

Luan D, Szlam F, Tanaka KA, Barie PS, Varner J. Ensembles of uncertain mathematical models can identify network responses to therapeutic interventions. Mol Biosyst. 2010 Nov;6(11):2272-86. doi: 10.1039/b920693k. Epub 2010 Sep 16. PMID: 20844798.

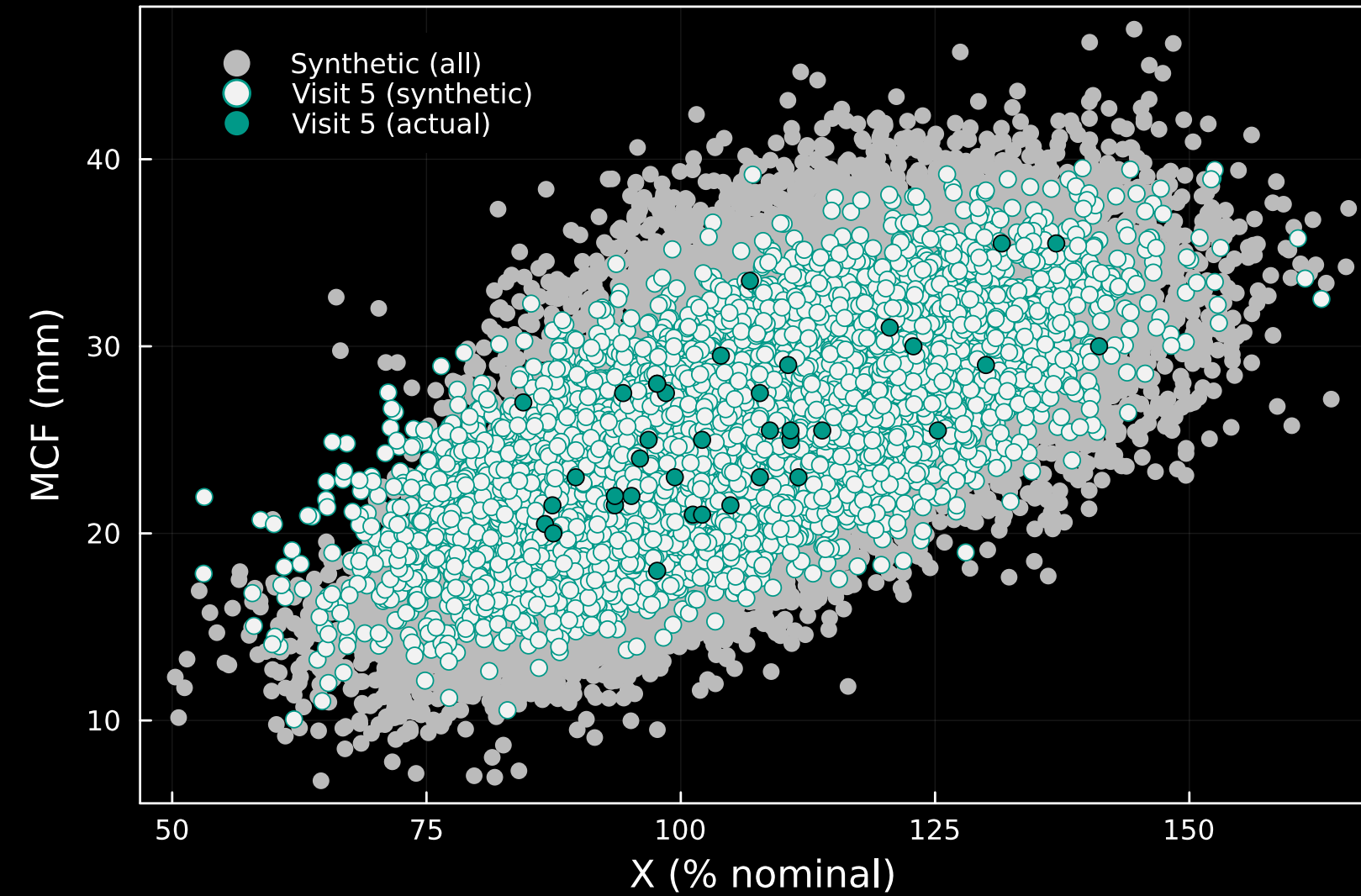
We constructed a synthetic population of patients from actual patients at V1 (basal), V5 (first trimester), and V6 (third trimester) for 32 clinical features (coagulation and fibrinolytic proteomes and functional data).

Joint multivariate normal distributions were constructed by computing the mean and covariance (numerical) from $N = 37$ patients (each with full V1, V5, and V6 data) conditioned on the visit index

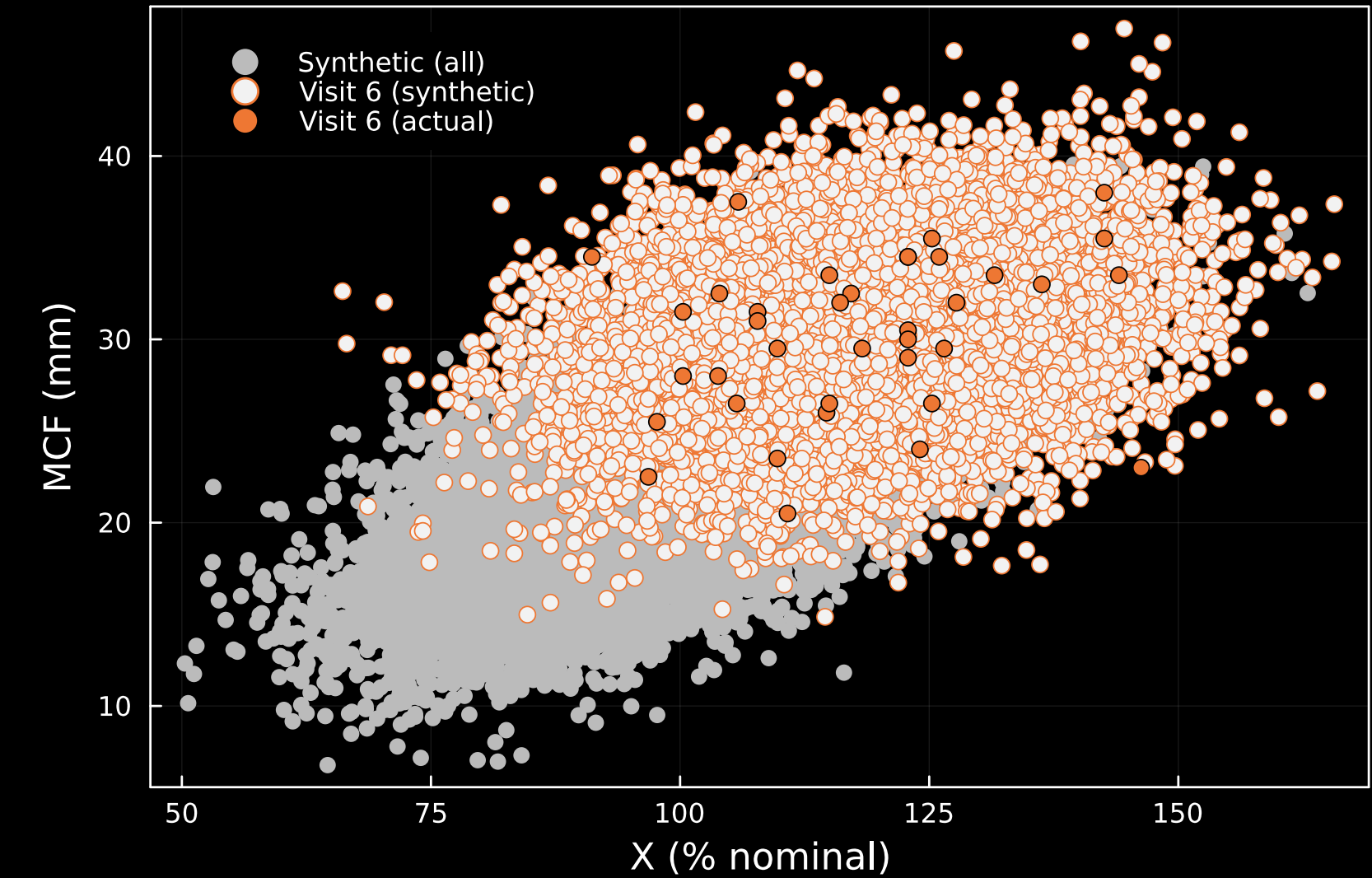
V1: not pregnant



V5: first trimester

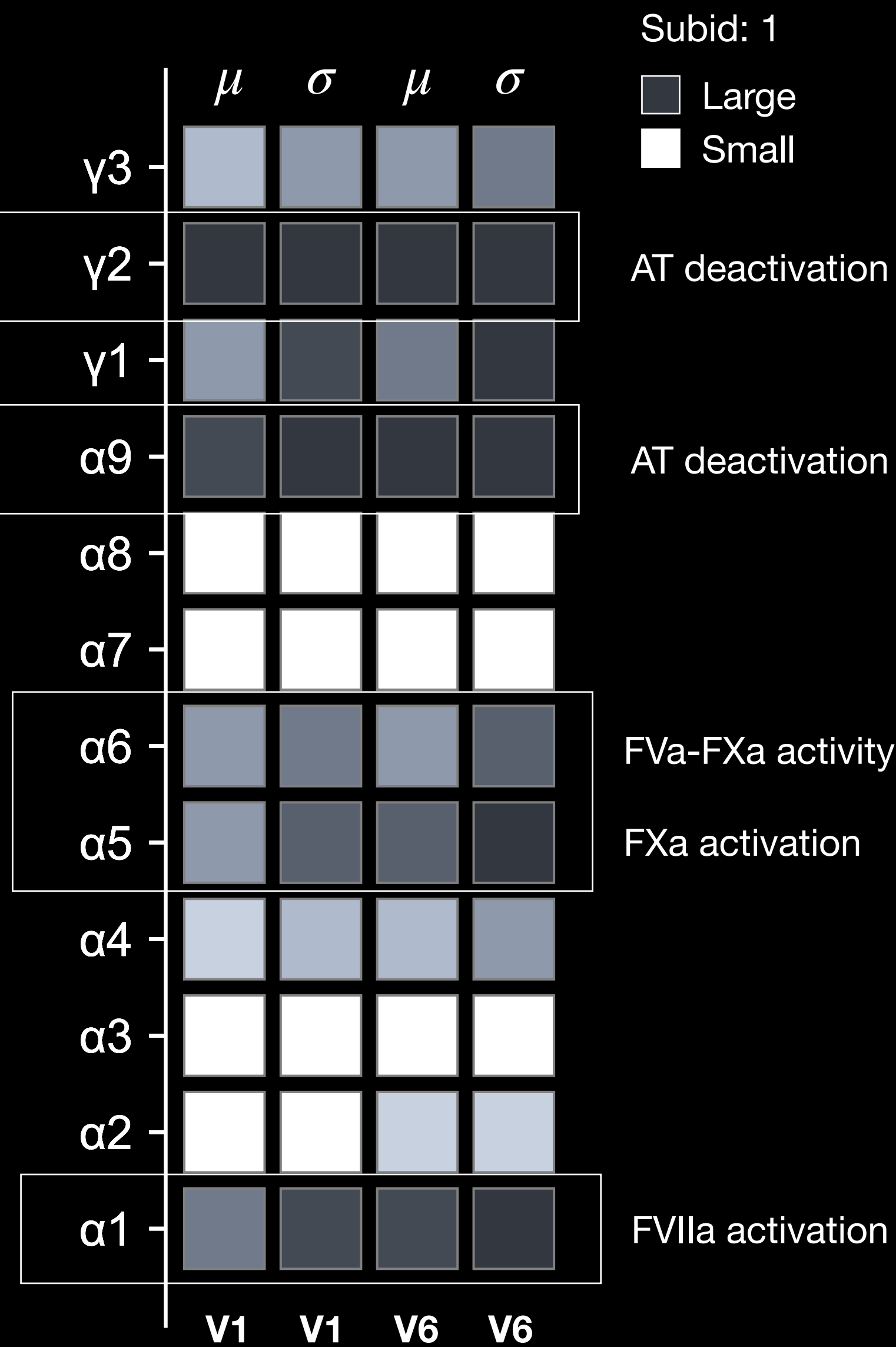
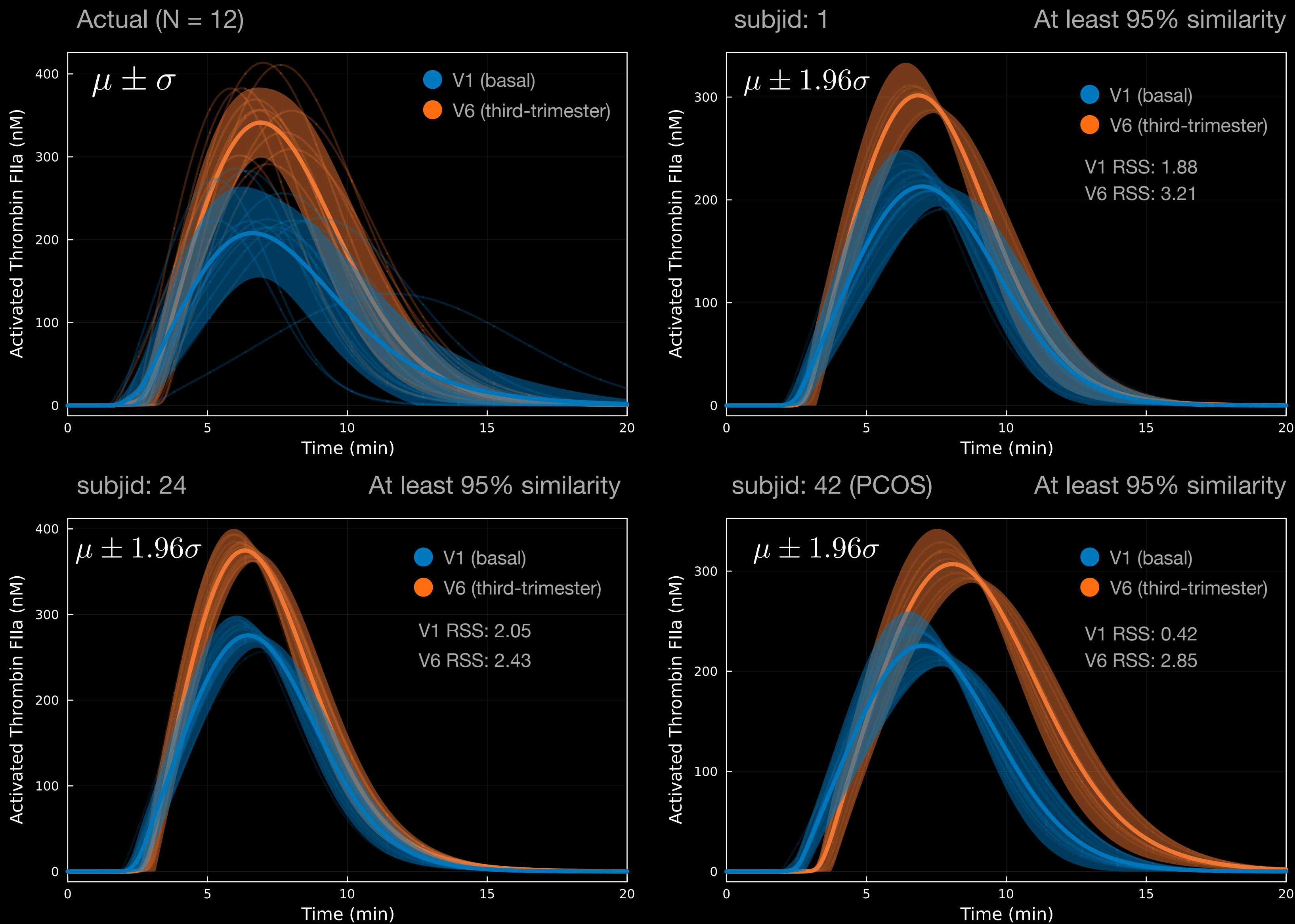


V6: third trimester



As the patient moves through the course of pregnancy, they move from the lower left to the upper right in the Factor X versus Maximum Clot Firmness (MCF) plane

We interrogated the synthetic and actual populations at the level of individual patients to better understand what factors were driving Thrombin and Fibrinolytic dynamics.



Summary and Acknowledgments

We developed (and used) the `BSTModelKit.jl` package to model Coagulation dynamics before and during pregnancy

- `BSTModelKit.jl` is a lightweight package that rides above `DifferentialEquations.jl`. The `BSTModelKit.jl` package transforms a model specification file into Biochemical Systems Theory (BST) S-system model

We interrogated synthetic and actual patient data at the level of individual patients to better understand what factors were driving Thrombin generation.

- Patients at V6 (third trimester) had a higher peak thrombin and a larger area under the curve relative to V1 (basal). These features were recapitulated in the synthetic population.

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- The Interaction of Basal Risk, Pharmacological Ovulation Induction, Pregnancy and Delivery on Hemostatic Balance NIH NHLBI R-33 HL 141787 (PI's Bernstein, Orfeo)
- The Pregnancy Phenotype and Predisposition to Preeclampsia NIH NHLBI R01 HL 71944 (PI Bernstein)

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—— Questions? ———

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